

# PENETRATION TESTING ASSESSMENT REPORT

Nmap Network Reconnaissance — Metasploitable Lab

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|                   |                                                                     |
|-------------------|---------------------------------------------------------------------|
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| Classification:   | Educational / Controlled Lab Environment                            |

## EXECUTIVE SUMMARY

A structured, flag-based Nmap reconnaissance challenge was completed against a Metasploitable target within an isolated VirtualBox lab. The exercise progressed from basic host discovery through service/OS fingerprinting, XML reporting, firewall evasion, and NSE-based web enumeration, validating each stage with a captured flag. 9 of 10 tasks are documented in this report.

## SCOPE

|                    |                                          |
|--------------------|------------------------------------------|
| Target IP Address: | 192.168.88.20 / 192.168.88.253           |
| Target System:     | Metasploitable (isolated VirtualBox lab) |
| Attacker Host:     | Kali Linux                               |
| Tools Used:        | Nmap, Wireshark, Kali Linux              |

## Task 1:

Basic host discovery

```
root@kali: /home/kali
Session Actions Edit View Help
(root@kali)-[/home/kali]
# nmap -sn 192.168.88.20
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-05 00:23 -0500
Nmap scan report for 192.168.88.20
Host is up (0.094s latency).
MAC Address: FC:F8:AE:D4:E9:2D (Intel Corporate)
Nmap done: 1 IP address (1 host up) scanned in 0.70 seconds
```

CTF { host - up }

## Task 2:

Identify Web services

```
(root@kali)-[/home/kali]
# nmap -p 21, 80, 443 192.168.88.20
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-05 00:25 -0500
Failed to resolve "80,".
Nmap scan report for 192.168.88.20
Host is up (0.079s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
MAC Address: FC:F8:AE:D4:E9:2D (Intel Corporate)
Nmap done: 2 IP addresses (1 host up) scanned in 5.76 seconds
```

CTF { Open\_port: 21/tcp }

### Task 3:

Full port sweep

```
(root@kali)-[/home/kali]
# nmap -p- 192.168.88.20
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-05 00:27 -0500
Nmap scan report for 192.168.88.20
Host is up (0.016s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
39837/tcp open  unknown
44440/tcp open  unknown
46088/tcp open  unknown
54154/tcp open  unknown
MAC Address: FC:F8:AE:D4:E9:2D (Intel Corporate)

Nmap done: 1 IP address (1 host up) scanned in 38.22 seconds
```

CTF { HIDDEN\_PORT: 8180, 39837, 44440, 46088, 54154 }

#### Task 4:

#### Service version detection

```
(root@kali)-[/home/kali]
└─$ nmap -sV 192.168.88.20
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-05 00:35 -0500
Nmap scan report for 192.168.88.20
Host is up (0.015s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: 8WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: FC:F8:AE:D4:E9:2D (Intel Corporate)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.45 seconds
```

CTF { SERVICE\_VERSION: Apache Tomcat/Coyote JSP engine 1.1 }

## Task 5:

### OS Fingerprinting

```
(root@kali)-[/home/kali]
# nmap -O 192.168.88.20
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-05 00:41 -0500
Nmap scan report for 192.168.88.20
Host is up (0.030s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: FC:F8:AE:D4:E9:2D (Intel Corporate)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.32 seconds
```

CTF: { SERVICE\_VERSION: linux 2.6.9 - 2.6.33 }

## Task 6:

Aggressive recon:

```
script results:
security-mode:
ount_used: <blank>
entication_level: user
allenge_response: supported
sage_signing: disabled (dangerous, but default)
t: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
s-discovery:
  Unix (Samba 3.0.20-Debian)
outer name: metasploitable
BIOS computer name:
ain name: localdomain
W: metasploitable.localdomain
tem time: 2026-03-06T00:45:50-05:00
-skew: mean: 1h14m55s, deviation: 2h30m00s, median: -5s
time: Protocol negotiation failed (SMB2)

UTE
  ADDRESS
2 ms 192.168.88.253
```

CTF: { host-name:metasploitable }

## Task 7:

Save result:

```
l-$ cat m-adeel-tariq.xml
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE nmaprun>
<?xml-stylesheet href="file:///usr/share/nmap/nmap.xsl" type="text/xsl"?>
<!-- Nmap 7.98 scan initiated Fri Mar 6 00:50:30 2026 as: /usr/lib/nmap/nmap -6#45;privileged -sV -O -T4 -oX m-adeel-tariq.xml 192.168.88.253 -->
<nmaprun scanner="nmap" args="/usr/lib/nmap/nmap -6#45;privileged -sV -O -T4 -oX m-adeel-tariq.xml 192.168.88.253" start="1772776230" startstr="Fri Mar 6 00:50:30 2026" version="7.98" xmloutputversion="1.05">
  <scaninfo type="syn" protocol="tcp" numservices="1000" services="1,3-4,6-7,9,13,17,19-26,30,32-33,37,42-43,49,53,70,79-85,88-90,99-100,106,109-111,113,119,125,135,139,143-144,146,161,163,179,199,211-212,222,254-256,259,264,280,301,306,311,340,366,389,406-407,416-417,425,427,443-445,458,464-465,481,497,500,512-515,524,541,543-545,548,554-555,563,587,593,616-617,625,631,636,646,648,666-668,683,687,691,700,705,711,714,720,722,726,749,765,777,783,787,800-801,808,843,873,880,888,898,900-903,911-912,981,987,990,992-993,995,999-1002,1007,1009-1011,1021-1100,1102,1104-1108,1110-1114,1117,1119,1121-1124,1126,1130-1132,1137-1138,1141,1145,1147-1149,1151-1152,1154,1163-1166,1169,1174-1175,1183,1185-1187,1192,1198-1199,1201,1213,1216-1218,1233-1234,1236,1244,1247-1248,1259,1271-1272,1277,1287,1296,1300-1301,1309-1311,1322,1328,1334,1350,1417,1433-1434,1443,1455,1461,1494,1500-1501,1503,1521,1524,1533,1556,1580,1583,1594,1600,1641,1658,1666,1687-1688,1700,1717-1721,1723,1755,1761,1782-1783,1801,1805,1812,1839-1840,1862-1864,1875,1900,1914,1935,1947,1971-1972,1974,1984,1998-2010,2013,2020-2022,2030,2033-2035,2038,2040-2043,2045-2049,2065,2068,2099-2100,2103,2105-2107,2111,2119,2121,2126,2135,2144,2160-2161,2170,2179,2190-2191,2196,2200,2222,2251,2260,2288,2301,2323,2366,2381-2383,2393-2394,2399,2401,2492,2500,2522,2525,2557,2601-2602,2604-2605,2607-2608,2638,2701-2702,2710,2717-2718,2725,2800,2809,2811,2869,2875,2909-2910,2920,2967-2968,2998,3000-3001,3003,3005-3006,3011,3017,3030-3031,3052,3071,3077,3128,3168,3211,3221,3260-3261,3268-3269,3283,3300-3301,3306,3322-3325,3333,3351,3367,3369-3372,3389-3390,3404,3476,3493,3517,3527,3546,3551,3580,3659,3689-3691,3703,3737,3766,3784,3800-3801,3809,3814,3826-3828,3851,3869,3871,3878,3880,3889,3905,3914,3918,3920,3945,3971,3986,3995,3998,4000-4006,4045,4111,4125-4126,4129,4224,4242,4279,4321,4343,4443-4446,4449,4550,4567,4662,4848,4899-4900,4998,5000-5004,5009,5030,5033,5050-5051,5054,5060-5061,5080,5087,5100-5102,5120,5190,5200,5214,5221-5222,5225-5226,5269,5280,5298,5357,5405,5414,5431-5432,5440,5500,5510,5544,5550,5555,5560,5566,5631,5633,5666,5678-5679,5718,5730,5800-5802,5810-5811,5815,5822,5825,5850,5859,5862,5877,5900-5904,5906-5907,5910-5911,5915,5922,5925,5950,5952,5959-5963,5985-5989,5998-6007,6009,6025,6059,6100-6101,6106,6112,6123,6129,6156,6346,6389,6502,6510,6543,6547,6565-6567,6580,6646,6666-6669,6689,6692,6699,6779,6788-6789,6792,6839,6881,6901,6969,7000-7002,7004,7007,7019,7025,7070,7100,7103,7106,7200-7201,7402,7435,7443,7496,7512,7625,7627,7676,7741,7777-7778,7800,7911,7920-7921,7937-7938,7999-8002,8007-8011,8021-8022,8031,8042,8045,8080-8090,8093,8099-8100,8180-8181,8192-8194,8200,8222,8254,8290-8292,8300,8333,8383,8400,8402,8443,8500,8600,8649,8651-8652,8654,8701,8800,8873,8888,8899,8994,9000-9003,9009-9011,9040,9050,9071,9080-9081,9090-9091,9099-9103,9110-9111,9200,9207,9220,9290,9415,9418,9485,9500,9502-9503,9535,9575,9593-9595,9618,9666,9876-9878,9890,9900,9917,9929,9943-9944,9968,9998-10004,10009-10100,10012,10024-10025,10082,10180,10215,10243,10566,10616-10617,10621,10626,10628-10629,10778,11110-11111,11967,12000,12174,12265,12345,13456,13722,13782-13783,14000,14238,14441-14442,15000,15002-15004,15660,15742,16000-16001,16012,16016,16018,16080,16113,16992-16993,17877,17988,18040,18101,18988,19101,19283,19315,19350,19780,19801,19842,20000,20005,20031,20221-20222,20828,21571,22939,23502,24444,24800,25734-25735,26214,27000,27352-27353,27355-27356,27715,28201,30000,30718,30951,31038,31337,32768,32785,33354,33899,34571-34573,35500,38292,40193,40911,41511,42510,44176,44442-44443,44501,45100,48080,49152-49161,49163,49165,49167,49175-49176,49400,49999-50003,50005,50300,50389,50500,50636,50800,51103,51493,52673,52822,52848,52869,54045,54328,55055-55056,55555,55600,56737-56738,57294,57797,58080,60020,60443,61532,61900,62078,63331,64623,64680,65000,65129,65389"/>
  <verbose level="0"/>
  <debugging level="0"/>
  <hosthint><status state="up" reason="arp-response" reason_ttl="0"/>
  <address addr="192.168.88.253" addrtype="ipv4"/>
  <address addr="08:00:27:5F:85:7A" addrtype="mac" vendor="Oracle VirtualBox virtual NIC"/>
  <hostnames>
    </hostnames>
  </hosthint>
  <host starttime="1772776232" endtime="1772776254"><status state="up" reason="arp-response" reason_ttl="0"/>
  <address addr="192.168.88.253" addrtype="ipv4"/>
  <address addr="08:00:27:5F:85:7A" addrtype="mac" vendor="Oracle VirtualBox virtual NIC"/>
  <hostnames>
    </hostnames>
  </host>
</nmaprun>
```

CTF: { m-adeel-tariq.xml }

Task 9:  
Firewall evasion

```
(kali㉿kali)-[~]  
$ nmap -f 192.168.88.253  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-06 01:00 -0500  
Nmap scan report for 192.168.88.253  
Host is up (0.00044s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
MAC Address: 08:00:27:5F:85:7A (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 1.06 seconds
```

CTF: { EVADED: 8180 }



## Task 10:

### NSE Script Enumeration

NSE scri[pt] enumeration:

```
|_ http-server-header: Apache-Coyote/1.1
|_ http-enum:
|_ /admin/: Possible admin folder
|_ /admin/index.html: Possible admin folder
|_ /admin/login.html: Possible admin folder
|_ /admin/admin.html: Possible admin folder
|_ /admin/account.html: Possible admin folder
|_ /admin/admin_login.html: Possible admin folder
|_ /admin/home.html: Possible admin folder
|_ /admin/admin-login.html: Possible admin folder
|_ /admin/adminLogin.html: Possible admin folder
|_ /admin/controlpanel.html: Possible admin folder
|_ /admin/cp.html: Possible admin folder
|_ /admin/index.jsp: Possible admin folder
|_ /admin/login.jsp: Possible admin folder
|_ /admin/admin.jsp: Possible admin folder
|_ /admin/home.jsp: Possible admin folder
|_ /admin/controlpanel.jsp: Possible admin folder
|_ /admin/admin-login.jsp: Possible admin folder
|_ /admin/cp.jsp: Possible admin folder
|_ /admin/account.jsp: Possible admin folder
|_ /admin/admin_login.jsp: Possible admin folder
|_ /admin/adminLogin.jsp: Possible admin folder
|_ /manager/html/upload: Apache Tomcat (401 Unauthorized)
|_ /manager/html: Apache Tomcat (401 Unauthorized)
|_ /admin/view/javascript/fckeditor/editor/filemanager/connectors/test.html: OpenCart/FCKeditor File upload
|_ /admin/includes/FCKeditor/editor/filemanager/upload/test.html: ASP Simple Blog / FCKeditor File Upload
|_ /admin/jscript/upload.html: Lizard Cart/Remote File upload
|_ /webdav/: Potentially interesting folder
MAC Address: 08:00:27:5F:85:7A (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 37.38 seconds
```

## Key Findings

- The target exposed an unusually high number of open services (25) for a hardened system, consistent with an intentionally vulnerable lab host.
- Version detection revealed Apache Tomcat/Coyote 1.1 on a non-standard port (8180) and an outdated Linux 2.6.x kernel — both indicative of legacy, unpatched software.
- NSE http-enum scan surfaced multiple exposed admin login paths and a FCKeditor file-upload endpoint, a common vector for unauthenticated remote file upload attacks.
- Basic packet fragmentation (-f) was sufficient to bypass the target's filtering rules, highlighting the limits of simple firewall configurations against evasion techniques.

## What I Learned

Practical application of Nmap's full scanning lifecycle — from stealth discovery to aggressive service fingerprinting — and how output formats (XML) and evasion flags support both reporting and real-world reconnaissance workflows.

## Note

Task 8 was not captured in the source documentation for this exercise; results for Tasks 1–7, 9, and 10 (9 of 10) are documented in this report.

## Environment Disclaimer

All scanning was performed against an isolated, intentionally vulnerable lab target (Metasploitable) inside a controlled VirtualBox environment as part of structured academic training. No unauthorized systems were accessed.